

Cambridge A Level Chemistry Assessment

Limiting Reagent and Excess Reagent in Solution Reactions

Instructions:

Answer both questions in your assigned set. Total: 20 marks.

Show all working clearly. Correct units and appropriate significant figures are required.

Set A - Limiting Reagent and Excess Reagent Assessment

Name:	Class:	Date:
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Total: 20 marks. Answer both questions. Show working, units, and appropriate significant figures.

Question 1: Formulae and limiting reagent logic [5 marks]

(a) State the two formulae needed to calculate moles from solution data and mass from moles. Include the required unit for solution volume.	[2]
(b) Explain why the limiting reagent cannot always be identified by comparing the two raw mole values directly.	[2]
(c) Describe a reliable method for identifying the limiting reagent from a balanced equation.	[1]

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Question 2: Precipitate formation and excess reagent [15 marks]

Copper(II) sulfate solution is mixed with sodium hydroxide solution. A blue precipitate of copper(II) hydroxide forms.

Balanced equation: $\text{CuSO}_4(\text{aq}) + 2\text{NaOH}(\text{aq}) \rightarrow \text{Cu}(\text{OH})_2(\text{s}) + \text{Na}_2\text{SO}_4(\text{aq})$

Solutions mixed: 25.0 cm³ of 0.200 mol dm⁻³ CuSO₄(aq) and 30.0 cm³ of 0.300 mol dm⁻³ NaOH(aq).

Question step	Marks
(a) Convert each volume from cm ³ to dm ³ , then calculate the initial moles of each reactant.	[4]
(b) Use the balanced equation to determine which reactant is limiting and which reactant is in excess. Show your comparison clearly.	[3]
(c) Calculate the amount, in moles, of Cu(OH) ₂ precipitate formed.	[2]
(d) Calculate the mass, in grams, of Cu(OH) ₂ precipitate formed.	[3]
(e) Calculate the amount, in moles, of the excess reactant remaining after the reaction is complete.	[2]
(f) Calculate the mass of the excess reactant remaining after the reaction is complete.	[1]

Total for this set: _____ / 20

Set B - Limiting Reagent and Excess Reagent Assessment

Name:	Class:	Date:
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Total: 20 marks. Answer both questions. Show working, units, and appropriate significant figures.

Question 1: Formulae and limiting reagent logic [5 marks]

(a) Define the terms limiting reagent and excess reagent in the context of a precipitation reaction. [2 marks]	[2]
(b) State the formula used to calculate moles of a solute in a solution and explain why cm^3 must be converted to dm^3 . [2 marks]	[2]
(c) Explain how the coefficients in a balanced equation are used to decide which reagent is limiting. [1 mark]	[1]

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Question 2: Precipitate formation and excess reagent [15 marks]

Barium chloride solution is mixed with sodium sulfate solution. A white precipitate of barium sulfate forms.

Balanced equation: $\text{BaCl}_2(\text{aq}) + \text{Na}_2\text{SO}_4(\text{aq}) \rightarrow \text{BaSO}_4(\text{s}) + 2\text{NaCl}(\text{aq})$

Solutions mixed: 20.0 cm³ of 0.150 mol dm⁻³ BaCl₂(aq) and 25.0 cm³ of 0.100 mol dm⁻³ Na₂SO₄(aq).

Assume the reaction goes to completion and the volumes are additive.

Question step	Marks
(a) Convert each volume from cm ³ to dm ³ , then calculate the initial moles of each reactant.	[4]
(b) Use the balanced equation to determine which reactant is limiting and which reactant is in excess. Show your comparison clearly.	[3]
(c) Calculate the amount, in moles, of BaSO ₄ precipitate formed.	[2]
(d) Calculate the mass, in grams, of BaSO ₄ precipitate formed.	[3]
(e) Calculate the amount, in moles, of the excess reactant remaining after the reaction is complete.	[2]
(f) Calculate the final concentration, in mol dm ⁻³ , of the excess ion Ba ²⁺ (aq) remaining in the mixture.	[1]

Total for this set: _____ / 20